

Noise Cancelling Headphones

When your headphones are on, you want only music, and not your screaming neighbours! That's where noise isolation and cancellation comes in...

Passive Noise Cancelling

A typical in-ear headphone has a protruding portion that enters your ear canal and seals it from external noise. This design also helps transfer sound directly from the driver into your ear — that's why the different eartips for different ear sizes. They isolate sound at best, don't really cancel it all that well.



Active Noise Cancelling

Active noise cancellation at its very basic utilises a physics-based technique called 'destructive interference'; it, that creates a sound wave that is the direct inverse of the external noise, effectively removing it. Hardware noise filters are simple circuits, whereas software runs on a DSP or digital signal processor. A microphone placed externally on the headphone monitors external noise, and this noise is converted to a digital signal that superimposes it over the noise wave, effectively nullifying it.

Passive Noise Cancelling - Circumaural

The sealed enclosure of passive noise cancelling circumaural headphones coupled with earpads that cover your entire ear acting as a filter to weak external sounds. Some such headphones block external noise well while others are moderately effective. Such headphones are called noise-isolating headphones. This is because they (quite obviously) block out external sound waves rather than neutralise them.

Interference - Noise Wave

External noise, unwanted sound is received by the headphone via microphone.

Noise Cancellation - Wave Diagram

The red sound wave is received by the headphone with active noise cancellation. This wave is then nullified by a reverse wave (blue) output generated by the noise cancellation circuit present inside the headphone, both - at the left and the right ear.

Music - What We Want to Hear

While the external interference is being actively dealt with by the headphone's noise cancellation circuit, music or sound we want to hear is passed "as is" without any tampering, much to the listener's joy.